

Lab Report : 06

The for Loop in C++

Objective

To understand the working of the for loop in C++ and demonstrate its use in performing repetitive tasks such as printing sequences, calculating sums, and controlling iterations.

Theory

- A **loop** is a control structure that allows repeated execution of a block of code.
- The for **loop** is an **entry-controlled loop** in C++.
- Syntax:

```
for (initialization; condition; update) {  
    // loop body  
}
```

- **Initialization:** executed once at the beginning.
- **Condition:** checked before each iteration; if false, loop terminates.
- **Update:** modifies the loop variable after each iteration.

Algorithm

1. Start the program.
2. Initialize the loop variable.
3. Check the condition: if true, execute the loop body; if false, exit.
4. Perform the update expression.
5. Repeat steps 3–4 until the condition becomes false.
6. End the program.

Programs & Outputs

Example 1: Print Numbers 1–5

```
#include <iostream> // Input output library ko include kiya gaya hai
using namespace std; // Standard namespace use kiya gaya hai

int main() { // Main function jahan se program start hota hai

    int sum = 0; // Sum variable banaya aur uski initial value 0 rakhi

    // For loop 1 se 10 tak chalega
    for (int i = 1; i <= 10; i++) {
        sum += i; // Har iteration me i ki value sum me add hogi
    }

    cout << "Sum = " << sum; // Final sum screen par display hoga

    return 0; // Program successfully end hua
}
```

Output:

Code

1 2 3 4 5

Example 2: Sum of First 10 Natural Numbers

```
#include <iostream> // Input aur output ke liye library include ki gayi hai
using namespace std; // Standard namespace use kiya gaya hai

int main() { // Program ka main function
    int sum = 0; // Sum naam ka variable banaya aur uski value 0 rakhi

    // Loop 1 se 10 tak chalega
    for (int i = 1; i <= 10; i++) {
        sum += i; // Har number ko sum me add kiya ja raha hai
    }

    cout << "Sum = " << sum; // Final sum display kar raha hai

    return 0; // Program successful end hua
}
```

Output:

Sum = 55

Example 3: Countdown

```
#include <iostream> // Input aur output ke liye library include ki gayi hai
using namespace std; // Standard namespace use kiya gaya hai

int main() { // Main function jahan se program start hota hai

    // For loop 5 se start hoga aur 1 tak chalega
    for (int i = 5; i > 0; i--) {

        cout << i << " "; // Har value ko screen par print kiya ja raha hai
    }

    return 0; // Program successfully end hua
}
```

Output:

Code

5 4 3 2 1

Results

- The for loop successfully executes repetitive tasks with concise syntax.
- It is most effective when the **number of iterations is known in advance**.
- Different variations (incrementing, decrementing, skipping values) can be applied depending on the problem.

Conclusion

The for loop in C++ is a powerful construct for iteration. It allows initialization, condition checking, and updating in a single line, making programs more readable and efficient. It is widely used in scenarios like printing sequences, summing numbers, and performing repetitive calculations.

Lab Report Tasks:

Task (1):

For Loop

- Write a C++ program that uses a while loop to display numbers from 1 to 10.
- Display each number on a separate line.

Code:

```
#include <iostream> // Input aur output ke liye library
using namespace std; // Standard namespace use karne ke liye

int main() {
    int i = 1; // Counter variable ko 1 se initialize kiya

    // Jab tak condition (i <= 10) true rahegi, loop chalega
    while (i <= 10) {
        cout << i << endl; // i ki current value print karega
        i++;                // i ki value ko 1 se increase karega
    }

    return 0; // Program ka end
}
```

Task (2):

Nested for Loop : Multiplication Table

- Write a C++ program that generates a multiplication table from 1 to 10.
- Use nested for loops to iterate through the rows and columns of the table.

- Display the multiplication table in a formatted manner

Code:

```
#include <iostream> // Input aur output ke liye library
using namespace std; // Standard namespace use karne ke liye

int main() {
    // Rows ke liye outer loop (1 se 10 tak)
    for (int i = 1; i <= 10; i++) {

        // Columns ke liye inner loop (1 se 10 tak)
        for (int j = 1; j <= 10; j++) {

            cout << i * j << "\t"; // i aur j ka multiplication print karega with tab spacing
        }

        cout << endl; // Har row ke baad next line me move karega
    }

    return 0; // Program ka end
}
```

Task (3):

Sum using while Loop

- Write a C++ program that prompts the user to enter a positive integer.
- Use a for loop to calculate the sum of all numbers from 1 to the entered
- integer.
- Display the calculated sum.

Code:

```

#include <iostream> // Input aur output ke liye library
using namespace std; // Standard namespace use karne ke liye

int main() {
    int num, sum = 0; // Variables declare kiye, sum ko 0 se initialize kiya

    cout << "Enter a positive integer: "; // User se positive integer input lene ke liye message
    cin >> num; // User se input lena

    // For loop 1 se num tak chalega
    for (int i = 1; i <= num; i++) {
        sum += i; // i ki value ko sum me add karega
    }

    cout << "Sum = " << sum << endl; // Final sum display karega

    return 0; // Program ka end
}

```

Task (4):

- **Nested Pattern Printing using nested for loop**
- Write a C++ program that uses nested for loops to print a pattern of asterisks.
- The pattern should form a triangle shape, with each row having an increasing number of asterisks.
- The number of rows in the pattern should be determined by user input.

Code:

```
#include <iostream> // Input aur output ke liye library
using namespace std; // Standard namespace use karne ke liye

int main() {
    int rows; // Triangle ki rows ki quantity store karne ke liye variable

    cout << "Enter number of rows: "; // User se rows ka number lene ke liye message
    cin >> rows; // User se input lena

    // Outer loop rows ki quantity control karega
    for (int i = 1; i <= rows; i++) {
        // Inner loop har row me stars print karega
        for (int j = 1; j <= i; j++) {
            cout << "* "; // Star ko space ke sath print karega
        }

        cout << endl; // Har row ke baad next line me move karega
    }

    return 0; // Program ka end
}
```